Trust and Age Bayesian Digital Twins: Provable Predictive Maintenance under Industrial Network Impairments

Yassir Ameen Al-Karawi, Hamed Al-Raweshidy

***Abstract*—Predictive-maintenance digital twins often assume synchronous, equally reliable sensor streams, although industrial packets may be delayed, reordered, or lost. This paper introduces TABDT, a trust and age Bayesian twin for remaining useful life (RUL) prediction under geometrically distributed packet delays and a finite deadline. Received observations are advanced to the current decision time and weighted by age and sensor precision before entering a random-effects Wiener model. For a known-drift, single-sensor scalar anchor, we derive a closed-form steady-state variance bound, identify two RUL-error scaling regimes, and give sufficient conditions for a unique network-dependent maintenance limit. This guarantee is separate from TABDT's approximate delayed-data recursion. Across 4 000 known-drift anchor paths, the bound is within 1.3% of empirical variance. In a separate 400-unit study, TABDT reduces RUL root-mean-square error (RMSE) by up to 27.9%. On all 15 XJTU-SY run-to-failure bearings, normalized RUL RMSE is 47.70 points, compared with 52.11 for stale-as-fresh fusion and 53.02 for discard-stale filtering. Simulated 90% interval coverage is 93.4–95.6%; external coverage is 76.9%, indicating model mismatch rather than field calibration.**

***Index Terms*—Digital twin, predictive maintenance, industrial cyber-physical systems, remaining useful life, Kalman filtering, age of information, uncertainty quantification, Wiener process.**

I. Introduction

INDUSTRIAL cyber-physical systems (ICPS) combine sensing, communication, computation, and control. Digital twins (DTs) support predictive maintenance (PdM) by fusing condition data with degradation models to estimate remaining useful life (RUL) [1], [2], [3]. Recent approaches often assume synchronized, homogeneous observations [4], [5], although trustworthy deployment depends on input quality and traceability [2].

Plant networks violate these assumptions through delayed, reordered, and lost packets and heterogeneous sensor calibration. Treating stale data as current biases the state estimate and understates uncertainty. Existing network-aware DTs [6], uncertainty guidance [7], [8], intermittent filtering [9], AoI [10], CBM codesign [11], and robust fusion [12] do not jointly connect network impairment to first-passage prognosis. We address *(G1)* a scoped network-to-RUL certificate, *(G2)* principled use of delayed measurements, and *(G3)* explicit sensor credibility.

This work makes four contributions. *C1)* It couples a random-effects Wiener model to geometrically distributed packet delay and a finite deadline. *C2)* TABDT advances a delayed observation by the estimated degradation in transit and weights it by $w_s(a) = \tau_s/(1 + \tau_s aQ/R)$, separating sensor precision from age uncertainty. *C3)* For a known-drift, single-sensor scalar anchor, Theorem 1, Corollary 1, and Proposition 1 establish the variance bound, its two RUL-error regimes, and conditional properties of the maintenance limit; these guarantees do not extend to the approximate delayed-data recursion. *C4)* A 4 000-path anchor experiment evaluates the bound; a separate 400-unit common-random-number study and an XJTU-SY packet-impairment test quantify point error and interval coverage. Code and raw results are at https://github.com/YassirALKarawi/tabdt-reproducibility and permanently archived as Zenodo release v19, DOI: 10.5281/zenodo.21861066.

II. Related Work

A. Digital twins for predictive maintenance: DT-based bearing diagnosis, cross-domain transfer, and state-space health assessment establish the value of virtual replicas [13], [4], [5], while the ICPS prognostics literature centers RUL and maintenance decisions [1]. Random-effects Wiener models retain inverse Gaussian RUL, unit heterogeneity, and analytical tractability [14], [15], [16]; however, these works do not propagate packet age and sensor credibility into the RUL posterior.

B. Imperfect networks and information aging: Intermittent filtering, AoI, OOSM methods, and asynchronous Bayesian filtering address loss, freshness, delayed updates, and source-specific noise, respectively [9], [10], [17], [18], [19]. Bursty fading also shows why mean reception probability alone is insufficient [20]. These methods do not jointly connect network quality to first-passage RUL and asymmetric CBM costs [21]. Table I positions TABDT as a link between these models, not a new degradation law or exact OOSM replacement.

III. System Model and Problem Statement

A. Degradation Process

Consider a fleet of nominally identical assets, using a gas-compressor bearing as the running example. The health index

Y. A. Al-Karawi is with the Department of Communications Engineering, College of Engineering, University of Diyala, Baqubah, Diyala, Iraq (e-mail: yassir.alkarawi@uodiyala.edu.iq).

H. Al-Raweshidy is with the Wireless Networks and Communications Centre, Department of Electronic and Electrical Engineering, College of Engineering, Design and Physical Sciences, Brunel University London, Uxbridge UB8 3PH, U.K. (e-mail: hamed.al-raweshidy@brunel.ac.uk). Corresponding author: Hamed Al-Raweshidy.

TABLE I
POSITIONING RELATIVE TO PRIOR WORK

Study / model	Primary prognostic role	Network treatment	Uncertainty and sensor credibility	Decision or guarantee
State-space DT health assessment [5]	Synchronized virtual replica and predictive health assessment	Data acquisition in real time is assumed	Model and plant similarity without age-dependent sensor credibility	Experimental condition monitoring without an anchor network-to-RUL error law
DT transfer diagnosis [4], [13]	Training data generated by a DT and fault diagnosis across domains	Packet delay, reordering, and deadline loss are not modeled	Domain discrepancy is learned without a provenance score τ_s or stale-packet covariance	Empirical diagnosis without a control limit informed by the network
Adaptive Wiener RUL [16]	Stochastic degradation and online RUL prediction with adaptive drift	Measurements are available to the estimator without an explicit network	Random degradation and parameter uncertainty with a homogeneous observation channel	Probabilistic RUL without a synchronization certificate or PM rule
Intermittent Kalman filtering [9]	State estimation over lossy links	Bernoulli packet arrival or loss	Error-covariance analysis for intermittent observations	Stability and covariance theory without first-passage RUL or maintenance economics
AoI and OOSM methods [10], [17]	Quantify information age or update delayed tracking measurements	Explicit information age and arrival outside sequence	State update that accounts for delay while sensor provenance is outside scope	Timeliness or tracking correction without a DT and PdM decision chain
TABDT (this work)	**Bayesian RUL prediction and preventive maintenance**	**Geometric age, reordering, and finite delivery deadline**	**Joint trust and age weight $w_s(a)$ with decomposed RUL variance $\sigma^2_{\mathrm{RUL},t}$**	**Closed-form $\bar{P}(p)$, two scaling regimes, with $\xi^*(p)$ unique and monotone under the respective conditions of Proposition 1**

(HI) $x(t) \in \mathbb{R}$ increases with degradation and follows a Wiener process with a unit-specific drift,

$$x(t) = x_0 + \mu t + \sigma_B B(t), \qquad \mu \sim \mathcal{TN}_{[\mu_{\min},\infty)}(\bar{\mu}, \sigma^2_{\mu 0}), \qquad \mu_{\min} > 0. \tag{1}$$

Here $B(t)$ is standard Brownian motion, σ_B is the diffusion coefficient, and $\mathcal{TN}$ is a lower-truncated normal distribution. Its arguments denote the mean and variance of the untruncated parent normal. The positive support gives a proper first-passage model, while μ represents manufacturing and loading differences across units [14], [15], [16]. Failure occurs at the first crossing of the threshold D. Conditional on μ, the resulting time $T_f = \inf\{t : x(t) \geq D\}$ is inverse Gaussian (IG), with mean $(D - x_0)/\mu$ and shape $(D - x_0)^2/\sigma_B^2$ [14]. Hence the true RUL at time t is $\mathrm{RUL}_t = T_f - t$. Sampling every Δt and setting $x_k = x(k\Delta t)$ yields

$$\mathbf{z}_{k+1} = A\,\mathbf{z}_k + \mathbf{w}_k, \quad \mathbf{z}_k = \begin{bmatrix} x_k \\ \mu_k \end{bmatrix}, \quad A = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, \tag{2}$$

where $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathrm{diag}(Q, q_\mu))$ and $Q = \sigma_B^2 \Delta t$. The small random-walk variance q_μ allows the filter to accommodate slow changes in loading without discarding the unit-level drift model.

B. Sensing and Network Model

S condition-monitoring sensors observe the HI,

$$y_k^{(s)} = x_k + v_k^{(s)}, \qquad v_k^{(s)} \sim \mathcal{N}(0,\, R/\tau_s), \quad s = 1, \ldots, S. \tag{3}$$

Here R is nominal measurement variance and $\tau_s \in (0, 1]$ is a precision score; values below one inflate the variance. This calibration map is a TABDT modeling choice. From held-out innovations $r_i^{(s)}$, estimate $\widehat{V}_s = \max\{\epsilon_R, N_s^{-1} \sum_i [(r_i^{(s)})^2 - HP_i^- H^\top]\}$, where $H = [1\ \ 0]$, and set $\tau_s = \mathrm{clip}(R/\widehat{V}_s, \tau_{\min}, 1)$. Provenance may regularize this estimate, consistent with data-quality monitoring [7] and robust channel downweighting [12]. Here τ_s is fixed to isolate network effects from online trust estimation.

The packet carrying $y_k^{(s)}$ traverses the plant network with raw delay $A_k^{(s)}$ (equivalently, its age on arrival [10]),

$$A_k^{(s)} \sim \mathrm{Geom}_0(p), \qquad a_k^{(s)} = A_k^{(s)} \text{ if } A_k^{(s)} \leq a_{\max}, \tag{4}$$

and is discarded if the deadline is exceeded. We refer to $p = \mathbb{P}\{A = 0\}$ as the *timely synchronization probability*; the corresponding probability of deadline loss is $(1-p)^{a_{\max}+1}$. Analytical delays are packetwise independent, but unequal realizations can still reorder arrivals and produce an OOSM stream [17], [18].

Writing $q = 1 - p$ and $M = a_{\max}$, the delivered mass and its conditional mean age follow directly from the truncated geometric series:

$$P_{\mathrm{del}}(p) = \sum_{a=0}^{M} pq^a = 1 - q^{M+1}, \qquad \bar{a}_{\mathrm{del}}(p) = \mathbb{E}[A \mid A \leq M] = \frac{q[1 - (M+1)q^M + Mq^{M+1}]}{p(1 - q^{M+1})}. \tag{5}$$

Thus $1 - P_{\mathrm{del}}$ measures missing information, whereas $\bar{a}_{\mathrm{del}}$ measures the staleness of retained packets.

C. Problem Statement

At step k, the twin must use only arrived packets to report $\widehat{\mathrm{RUL}}_k$, a nominal $(1-\alpha)$ approximate prediction interval, and a preventive-maintenance (PM) trigger. We seek an accuracy guarantee for a tractable anchor model as a function of p and a fusion rule that does not overstate the information in stale or low-credibility observations. Table II summarizes the notation.

IV. THE TABDT FRAMEWORK

The architecture in Fig. 1 follows the path of a measurement from the physical asset to the maintenance decision. Its three central design choices—age compensation, joint trust–age weighting, and a network-informed PM limit—correspond to gaps G1–G3 identified above.

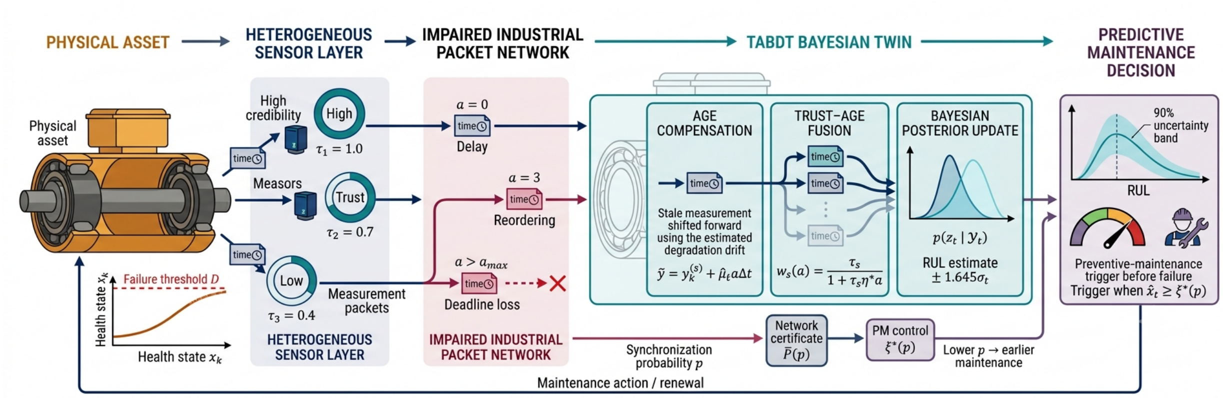

Fig. 1. End-to-end TABDT architecture. Timestamped sensor observations undergo delay, reordering, and deadline loss. Age compensation, trust–age fusion, and Bayesian updating yield state and RUL posteriors. The certificate $\bar{P}(p)$ provides the scale for the surrogate preventive-maintenance limit $\xi^*(p)$.

TABLE II
PRINCIPAL NOTATION

Symbol	Meaning
x_k, D	health index at step k, failure threshold
μ, $\bar{\mu}$, $\sigma_{\mu 0}$	unit drift, parent-normal mean and SD
σ_B, Q	diffusion coefficient and process variance $\sigma_B^2 \Delta t$
$y_k^{(s)}$, R, τ_s	measurement, nominal variance, sensor precision
$a_k^{(s)}$, p, q, $a_{\max}$	packet age, timely probability, $1-p$, deadline
$w_s(a)$, η^*	trust and age weight with $\eta^* = Q/R$
P_{del}	probability of delivery before the deadline
$\hat{\mathbf{z}}_k$, P_k	state estimate and error covariance
$\bar{P}(p)$	steady-state variance bound (Thm. 1)
ξ, ξ^*	PM control limit and its optimum (Prop. 1)
c_p, c_f	preventive / failure replacement cost

A. Age Compensation with Trust and Age Weighting

A packet $(y_k^{(s)}, k, s)$ arriving at step t with age $a = t - k$ measures the past state x_k rather than the current state. The two are related by $x_t = x_k + \mu a \Delta t + \nu_{k:t}$, where $\nu_{k:t} = \sum_{j=k+1}^{t} \varepsilon_j \sim \mathcal{N}(0, aQ)$. After shifting the packet by the estimated drift, its residual with respect to the current state is

$$\tilde{y} - H\mathbf{z}_t = v_k^{(s)} - \nu_{k:t} + a\Delta t(\hat{\mu}_t - \mu), \qquad H = [1\ 0]. \quad (6)$$

An exact OOSM retrodiction would retain both state–measurement covariance and the dependence between delayed packets [17], [18]. TABDT uses a cheaper forward correction: it holds the current drift fixed over the packet age and neglects correlations induced by shared process noise. Under this first-order conditional-independence approximation, (6) has marginal variance

$$R_{s,a}^{\mathrm{eff}} = \frac{R}{\tau_s} + aQ + (a\Delta t)^2 [P_t]_{\mu\mu}. \quad (7)$$

The estimator therefore advances the observation by the degradation missed in transit and assigns the retained marginal uncertainty to its likelihood:

$$\tilde{y} = y_k^{(s)} + \hat{\mu}_t \, a\Delta t, \quad (8)$$

$$\tilde{R} = \frac{R}{w} + (a\Delta t)^2 \, [P_t]_{\mu\mu}, \qquad w = w_s(a) = \frac{\tau_s}{1 + \tau_s \eta a}. \quad (9)$$

Using (8) and (9) sequentially does not recover the exact delayed-data posterior. This distinction is kept explicit, and the sensitivity, coverage, and exact-reprocessing experiments later quantify its practical consequence. For a fresh packet, τ_s controls the sensor variance; for a delayed packet, a discounts information further. Since $R/w_s(a) = R/\tau_s + \eta a R$, choosing

$$\eta^* = Q/R \quad (10)$$

makes $R/w_s(a) = R/\tau_s + aQ$: the nominal sensor contribution plus the Wiener variance accumulated during transit. The age rate is consequently the ratio of process to measurement variance, and it has the expected monotonicity

$$\frac{\partial w_s}{\partial a} = -\frac{\tau_s^2 \eta}{(1 + \tau_s \eta a)^2} < 0, \qquad \frac{\partial w_s}{\partial \tau_s} = \frac{1}{(1 + \tau_s \eta a)^2} > 0. \quad (11)$$

Section VI varies this ratio over four orders of magnitude. The remaining term in (9) accounts for uncertainty in the drift estimate; without it, an old packet would receive excessive weight precisely when the forward shift is least certain.

Finally, a delivered packet is fused once and only once. Reusing the last stored value in later updates would count the same observation repeatedly.

B. Bayesian Recursion

Equations (8) and (9) define the measurement updates in a standard sequential Kalman recursion for (2). Each step applies a time update followed by one scalar update per newly arrived packet, using $H = [1\ \ 0]$, $\tilde{y}$, and $\tilde{R}$. Algorithm 1 specifies packet order and single use. The prior $\hat{\mu}_0 = \bar{\mu}$, with variance $\sigma_{\mu 0}^2$, uses the parent-normal moments as a Gaussian approximation to the truncated prior; state augmentation learns the unit drift [16].

C. RUL Posterior

Given the posterior mean $\hat{\mathbf{z}}_t = [\hat{x}_t, \hat{\mu}_t]^\top$ and covariance P_t, define $m_t = \max(D - \hat{x}_t, \epsilon_x)$ and $\hat{\mu}_t^+ = \max(\hat{\mu}_t, \epsilon_\mu)$. The forecast holds the drift fixed beyond t. This is exact when $q_\mu = 0$ and is a local approximation for the slow random walk in (2). Under that approximation, the first-passage RUL $L_t = T_f - t$ has the inverse Gaussian density

$$f_{L_t | x_t, \mu}(\ell) = \frac{D - x_t}{\sqrt{2\pi \sigma_B^2 \ell^3}} \exp\left[-\frac{(D - x_t - \mu\ell)^2}{2\sigma_B^2 \ell} \right], \quad \ell > 0. \quad (12)$$

Algorithm 1 Trust and age Bayesian recursion at time t

Require: posterior $(\hat{\mathbf{z}}_{t-1}, P_{t-1})$, unused packets $\mathcal{A}_t$, $Q, q_\mu, R, \{\tau_s\}, \eta^*, D, \Delta t, \sigma_B, \alpha, \epsilon_x, \epsilon_\mu, \xi^*(p)$
Invariant: each packet is fused once, in nondecreasing age $a = t - k$ (newest first).

1: **Predict:** $\hat{\mathbf{z}} \leftarrow A\hat{\mathbf{z}}_{t-1}$, $P \leftarrow AP_{t-1}A^\top + \mathrm{diag}(Q, q_\mu)$.
2: Sort $\mathcal{A}_t$ by (a, s). **For each** $(y_k^{(s)}, k, s) \in \mathcal{A}_t$ **do**
3: $a \leftarrow t - k$ and $w \leftarrow \tau_s/(1 + \tau_s\eta^* a)$.
4: Let $\hat{\mu} = [\hat{\mathbf{z}}]_\mu$ and $P_{\mu\mu} = [P]_{\mu\mu}$; set $\tilde{y} \leftarrow y_k^{(s)} + \hat{\mu}a\Delta t$ and $\tilde{R} \leftarrow R/w + (a\Delta t)^2 P_{\mu\mu}$.
5: $K \leftarrow PH^\top(HPH^\top + \tilde{R})^{-1}$ and $e \leftarrow \tilde{y} - H\hat{\mathbf{z}}$.
6: $\hat{\mathbf{z}} \leftarrow \hat{\mathbf{z}} + Ke$, $J \leftarrow I - KH$, and $P \leftarrow JPJ^\top + K\tilde{R}K^\top$.
7: **End for**. Symmetrize $P \leftarrow (P + P^\top)/2$, then set $\hat{\mathbf{z}}_t \leftarrow \hat{\mathbf{z}}$ and $P_t \leftarrow P$.
8: Evaluate (13) through (16). Set $u_t \leftarrow \mathbf{1}\{\hat{x}_t \geq \xi^*(p)\}$.

Ensure: $\hat{\mathbf{z}}_t, P_t, \widehat{\mathrm{RUL}}_t \pm z_{1-\alpha/2}\sigma_{\mathrm{RUL},t}$, and PM flag u_t.

The conditional mean and variance of (12) are $(D - x_t)/\mu$ and $(D - x_t)\sigma_B^2/\mu^3$, respectively. Evaluating the mean at the posterior estimates gives the point predictor

$$\widehat{\mathrm{RUL}}_t = \frac{m_t}{\hat{\mu}_t^+}. \tag{13}$$

For $g(x, \mu) = (D - x)/\mu$, when $D - \hat{x}_t > \epsilon_x$ and $\hat{\mu}_t > \epsilon_\mu$ the safeguards are inactive and the gradient at the posterior mean is

$$\nabla g_t = \begin{bmatrix} -(\hat{\mu}_t^+)^{-1} & -m_t(\hat{\mu}_t^+)^{-2} \end{bmatrix}^\top. \tag{14}$$

If either safeguard is active, (14) is instead used as a stabilized plug-in expression. Applying the law of total variance and the first-order delta method gives

$$\mathrm{Var}(L_t \mid \mathcal{Y}_t) \approx \mathbb{E}[\mathrm{Var}(L_t \mid \mathbf{z}_t) \mid \mathcal{Y}_t] + \nabla g_t^\top P_t \nabla g_t. \tag{15}$$

Substituting (14) into (15) separates the aleatoric first-passage part and the epistemic state-estimation part [14]:

$$\sigma_{\mathrm{RUL},t}^2 \approx \underbrace{\frac{m_t\sigma_B^2}{(\hat{\mu}_t^+)^3}}_{\text{aleatoric (IG)}} + \underbrace{\frac{[P_t]_{xx}}{(\hat{\mu}_t^+)^2} + \frac{m_t^2[P_t]_{\mu\mu}}{(\hat{\mu}_t^+)^4} + \frac{2m_t[P_t]_{x\mu}}{(\hat{\mu}_t^+)^3}}_{\text{epistemic (delta method)}}. \tag{16}$$

The nominal interval is $\widehat{\mathrm{RUL}}_t \pm z_{1-\alpha/2}\sigma_{\mathrm{RUL},t}$. It should not be interpreted as an exact Bayesian credible interval: (16) inserts the expected aleatoric component and neglects higher posterior moments. Its coverage must therefore be established empirically [8]. The delta expansion does, however, retain the state–drift covariance. The result in the next section bounds the $[P_t]_{xx}$ contribution only for the stated anchor filter; it is not a bound on Algorithm 1 with unknown drift and approximate OOSM updates.

V. Theoretical Guarantees

A. A Closed-Form Variance Certificate

For analytical tractability, consider a unit-trust, known-drift, single-sensor scalar anchor that accepts fresh packets and omits delayed ones. This reference model is separate from the approximate OOSM recursion in Algorithm 1. Let P_k be its one-step-ahead prior variance. Because a fresh measurement is available with probability p at each step [9], P_k obeys

$$P_{k+1} = P_k + Q - \gamma_k \frac{P_k^2}{P_k + R}, \qquad \gamma_k \sim \mathrm{Bern}(p). \tag{17}$$

Theorem 1 (Steady-state variance bound)**.** *For the scalar health submodel induced by (2) under known drift, suppose γ_k in (17) is independent of P_k. Then, for any $p \in (0, 1]$,*

$$\limsup_{k\to\infty} \mathbb{E}[P_k] \leq \bar{P}(p) = \frac{Q + \sqrt{Q^2 + 4pQR}}{2p}. \tag{18}$$

Proof. Write $g(P) = P^2/(P + R)$. Since $g''(P) = 2R^2/(P + R)^3 > 0$, g is convex on $[0, \infty)$, and Jensen's inequality gives $\mathbb{E}[g(P_k)] \geq g(\mathbb{E}[P_k])$. Taking expectations of (17),

$$\mathbb{E}[P_{k+1}] = \mathbb{E}[P_k] + Q - p\,\mathbb{E}[g(P_k)] \leq f\big(\mathbb{E}[P_k]\big),$$

with $f(u) = u + Q - p\,g(u)$. Because $g'(u) = (u^2 + 2uR)/(u + R)^2 \in [0, 1)$, the derivative $f'(u) \in (1 - p, 1]$. Hence f is nondecreasing, and the comparison sequence $u_{k+1} = f(u_k)$, $u_0 = \mathbb{E}[P_0]$, dominates $\mathbb{E}[P_k]$ for all k. The fixed points of f solve $Q = p\,g(u)$, i.e., $pu^2 - Qu - QR = 0$, whose unique positive root is $\bar{P}(p)$ in (18). Since $f(u) - u = Q - p\,g(u)$ is positive below $\bar{P}(p)$ and negative above it, and f is monotone, $u_k \to \bar{P}(p)$ from any initialization, which yields the claim. $\square$

Implicit differentiation of $p\bar{P}^2 = Q(\bar{P} + R)$ gives

$$\frac{d\bar{P}}{dp} = -\frac{\bar{P}^2}{2p\bar{P} - Q} = -\frac{\bar{P}^2}{\sqrt{Q^2 + 4pQR}} < 0. \tag{19}$$

Equation (19) shows that the variance bound increases strictly as the synchronization probability decreases.

Corollary 1 (Two network scaling regimes)**.** *For the known-drift anchor filter, let $\delta = Q/(pR)$. If $\delta \to 0$ (equivalently, $p \gg Q/R$), then*

$$\begin{aligned} \bar{P}(p) &= \sqrt{\frac{QR}{p}}\,[1 + O(\sqrt{\delta})], \\ \sigma_{\mathrm{RUL}}^{\mathrm{epi}} &\lesssim \frac{(QR)^{1/4}}{\mu}p^{-1/4}[1 + O(\sqrt{\delta})]. \end{aligned} \tag{20}$$

In the extreme-loss limit $pR/Q \to 0$,

$$\bar{P}(p) = \frac{Q}{p} + R + O\left(\frac{pR^2}{Q}\right), \qquad \sigma_{\mathrm{RUL}}^{\mathrm{epi}} \lesssim \frac{\sqrt{Q}}{\mu}p^{-1/2}. \tag{21}$$

For these parameters, $Q/R = 9 \times 10^{-4}$ and $p \geq 0.02$, so $\delta \leq 0.045$ and (20) applies. Halving p then increases the leading epistemic RUL scale by $2^{1/4} - 1 \approx 18.9\%$. In contrast, (21) gives the extreme-loss RUL-error exponent $-1/2$. Because TABDT retains delayed packets, the bound is used only as a network-dependent scale, not a guarantee for the full recursion.

B. Maintenance Threshold Informed by the Network

We next use the certificate to parameterize a preventive-maintenance cost model. Under the classical control-limit policy [21], the asset is replaced preventively (cost c_p) when the *estimate* $\hat{x}$ crosses $\xi < D$. An undetected crossing of D before the trigger costs $c_f > c_p$. Let $e_k = x_k - \hat{x}_k$ denote the zero-mean anchor-filter error. An unnoticed failure requires e_k

to exceed the safety margin $m = D - \xi$. Theorem 1 gives the distribution-free Cantelli bound

$$\limsup_{k\to\infty} \mathbb{P}\{e_k \geq m\} \leq \frac{\bar{P}(p)}{\bar{P}(p)+m^2}, \qquad m > 0. \tag{22}$$

Variance alone does not imply a Gaussian tail. We therefore introduce the following smooth surrogate, whose accuracy can be checked by simulation:

$$C_G(\xi,p) \;\triangleq\; \left[c_p + (c_f - c_p)\,\bar{\Phi}\!\left(\frac{D-\xi}{\sqrt{\bar{P}(p)}}\right)\right]\frac{\mu}{\xi}, \tag{23}$$

where $\bar{\Phi}$ and ϕ denote the standard normal tail and density. The bracketed factor approximates expected replacement cost, while μ/ξ approximates the renewal rate. Unlike the Cantelli result in (22), C_G is a design surrogate rather than a probabilistic upper bound.

Proposition 1 (Unique control limit and network monotonicity)**.** *Let $s = \sqrt{\bar{P}(p)}$ and $d = c_f - c_p > 0$. If*

$$\frac{Dd}{s\sqrt{2\pi}} > \frac{c_f + c_p}{2}, \tag{24}$$

the surrogate (23) has a unique interior minimizer $\xi^* \in (0, D)$ characterized by

$$(c_f - c_p)\,\frac{\xi^*}{\sqrt{\bar{P}(p)}}\,\phi\!\left(\frac{D-\xi^*}{\sqrt{\bar{P}(p)}}\right) = c_p + (c_f - c_p)\,\bar{\Phi}\!\left(\frac{D-\xi^*}{\sqrt{\bar{P}(p)}}\right). \tag{25}$$

Let $r^* = (D-\xi^*)/s$. Wherever $(r^*)^2\xi^* > D$, the safety margin $D - \xi^*$ is strictly increasing in s and hence nonincreasing in p.

Proof. Write $h(\xi) = c_p + d\bar{\Phi}((D-\xi)/s)$, so $C'_G(\xi,p)$ has the sign of $F(\xi) = \xi h'(\xi) - h(\xi)$. Here $F(0) < 0$, condition (24) is exactly $F(D) > 0$, and $F'(\xi) = \xi h''(\xi) > 0$ for $0 < \xi < D$. Hence the root is unique and gives (25). Implicit differentiation of (25) gives $\mathrm{d}(D-\xi^*)/\mathrm{d}s = r^* - D/(r^*\xi^*)$, which is positive under the stated condition. Finally, s decreases with p by (18). □

The surrogate combines a Gaussian tail with a first-order renewal scale and does not model diffusion overshoot near D; both approximations are examined in Section VI. Let $F_p(\xi) = \xi h'(\xi) - h(\xi)$ be the stationary-point residual from the proof of Proposition 1. Under (24), F_p is strictly increasing, so a bracketed root solve is preferable to an unconstrained optimizer. Algorithm 2 implements this calculation and, under the stated condition, returns the unique stationary point. If the condition fails and $F_p(D) \leq 0$, the surrogate decreases throughout $(0, D)$; its strict-domain infimum occurs as $\xi \uparrow D$, and the implementation returns the ϵ-interior approximation $D - \epsilon$.

C. Computational Complexity

Let $m_t = |\mathcal{A}_t|$. One step of Algorithm 1 costs $O(n^3 + m_t n^2 + m_t \log m_t)$ operations and $O(n^2 + m_t)$ memory. In steady state $\mathbb{E}[m_t] = SP_{\mathrm{del}} \leq S$, whereas the deadline gives $m_t \leq S(a_{\max}+1)$. For the fixed state dimension $n = 2$, the expected per-step complexity is therefore $O(Sn^2)$, although a deadline-sized burst can be more expensive. Algorithm 2 requires $O(\log(D/\epsilon))$ root iterations. These operation counts exclude platform-dependent latency and energy consumption, which require implementation-level profiling.

Algorithm 2 PM threshold informed by the network

Require: $p, Q, R, D, \mu, c_p, c_f$ and root tolerance $0 < \epsilon < D$.

$\bar{P} \leftarrow (Q + \sqrt{Q^2 + 4pQR})/(2p)$, $s \leftarrow \sqrt{\bar{P}}$, and $d \leftarrow c_f - c_p$.
$h(\xi) \leftarrow c_p + d\bar{\Phi}((D-\xi)/s)$ and $F_p(\xi) \leftarrow \xi d\phi((D-\xi)/s)/s - h(\xi)$.
if $F_p(D) \leq 0$ **then** set $\xi^* \leftarrow D - \epsilon$ and evaluate $C_G(\xi^*, p)$; **return**.
else solve $F_p(\xi) = 0$ on $(0, D)$ by a safeguarded bracketed method (bisection/Brent) until the bracket width is at most ϵ.
Set ξ^* to the bracketed root. Evaluate $C_G(\xi^*, p)$ from (23).

Ensure: unique interior $\xi^*(p)$ when (24) holds; otherwise, an ϵ-interior approximation to the boundary infimum.

TABLE III
SIMULATION CONFIGURATION

Parameter	Value	Parameter	Value
Δt	1 h	D	10 HI
$\bar{\mu}$	0.02 HI/h	$\sigma_{\mu 0}$	0.004 HI/h
$\mu_{\min}$	0.01 HI/h	P_0	$\mathrm{diag}(0.25, \sigma^2_{\mu 0})$
σ_B	0.03 HI/$\sqrt{\mathrm{h}}$	Q	9×10^{-4}
R	1.0	(τ_1,τ_2,τ_3)	(1.0, 0.7, 0.4)
q_μ	10^{-7}	$\eta^* = Q/R$	9×10^{-4}
$a_{\max}$	100 steps	(c_p, c_f)	(1, 10)
Monte Carlo units	400	eval. window	$[0.2, 0.9]\,T_f$

VI. Numerical Study

A. Simulation Setup and Model Verification

Table III specifies a bearing with $\bar{\mu} = 0.02$ HI/h, nominal mean life 500 h, and three unequal-precision sensors. The range $p = 0.02$ to 1 spans a 13.0% deadline-loss case ($a_{\max} = 100$) to ideal delivery. Simulator verification over 20 000 fixed-drift paths gave failure-time mean and standard deviation 501.3 h and 33.6 h, versus IG values 500.0 h and 33.5 h.

B. Baselines and Metrics

All four estimators use the same degradation paths, measurement noise, and packet realizations. B1 uses only the degradation model and parent-normal reference drift; B2 applies every delivered packet as current; B3 accepts only on-time packets [9]; and TABDT follows Algorithm 1. RUL RMSE and nominal 90% prediction-interval coverage (PICP) are evaluated over 20–90% of each life. Common paths, Gaussian draws, and inverse-CDF delay uniforms are reused across p. Plots show descriptive bars equal to ±1.96 standard errors from unweighted unit-level metrics; because the centers are pooled point-time metrics, these are not confidence intervals for the plotted estimands. For $\mathcal{I} = \{(n,k) : 0.2T_f^{(n)} \leq k\Delta t \leq 0.9T_f^{(n)}\}$, the metrics are

$$\begin{aligned} e_{n,k} &= \widehat{\mathrm{RUL}}_{n,k} - \mathrm{RUL}_{n,k}, \\ \mathrm{RMSE}_{\mathrm{RUL}} &= \left[|\mathcal{I}|^{-1}\sum_{(n,k)\in\mathcal{I}} e^2_{n,k}\right]^{1/2}, \\ \mathrm{PICP}_{0.9} &= |\mathcal{I}|^{-1}\sum_{(n,k)\in\mathcal{I}} \mathbf{1}\{|e_{n,k}| \leq 1.645\,\sigma_{\mathrm{RUL},n,k}\}. \end{aligned} \tag{26}$$

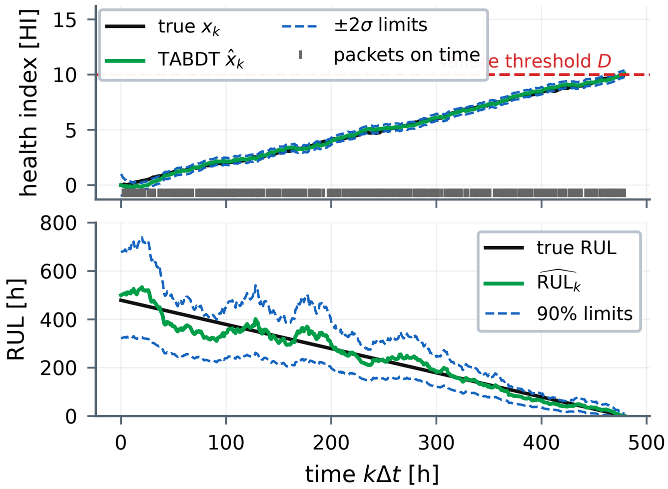

Fig. 2. One realization at $p = 0.4$. Top: true and estimated health index with paired $\pm 2\sigma$ limits, failure threshold D, and packets arriving on time (rug marks). Bottom: true RUL against the TABDT estimate (13) with paired 90% limits from (16).

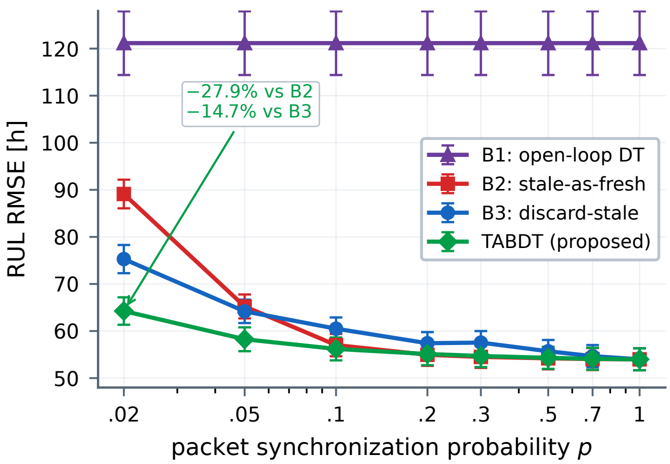

Fig. 3. RUL RMSE against timely synchronization probability p on a log scale. Bars are the descriptive quantities in Section VI-B; measurement-based methods coincide at $p = 1$.

C. Results

1) Representative trajectory: Fig. 2 shows a unit at $p = 0.4$. Timely packets arrive irregularly; prediction and subsequent updates progressively reflect its drift in the state estimate and the 90% RUL limits from (16).

2) RUL accuracy under network degradation: Figure 3 and Table IV give the main comparison. At $p = 1$, the measurement-based estimators coincide at 54.0 h; B1 remains at 121.2 h because it never learns unit drift. At $p = 0.02$, TABDT records 64.3 h versus 75.3 h for B3 and 89.1 h for B2, reductions of 14.7% and 27.9%. At $p = 0.05$, its 58.3 h RMSE is 9.2% and 10.7% lower, respectively. TABDT remains within 54.0–58.3 h for $p \geq 0.05$, so a twentyfold decrease from $p = 1$ to 0.05 adds 4.3 h in this experiment.

At $p = 0.02$, delivered packets have mean age 33.9 steps; treating them as current omits about $\bar{\mu}\,\mathbb{E}[a \mid a \leq a_{\max}]\Delta t = 0.68$ HI. TABDT restores this drift and includes $aQ + (a\Delta t)^2 P_{\mu\mu}$ in the variance, whereas B3 avoids bias by rejecting delayed information.

TABLE IV
HEADLINE COMPARISON AT THREE NETWORK CONDITIONS

Method	$p = 0.02$		$p = 0.05$		$p = 0.20$	
	RMSE [h]	PICP [%]	RMSE [h]	PICP [%]	RMSE [h]	PICP [%]
B1 open-loop	121.2	95.0	121.2	95.0	121.2	95.0
B2 stale-as-fresh	89.1	91.5	65.2	97.0	**55.0**	96.4
B3 discard-stale	75.3	95.7	64.2	95.8	57.4	95.9
TABDT (proposed)	**64.3**	93.4	**58.3**	94.5	55.1	95.2

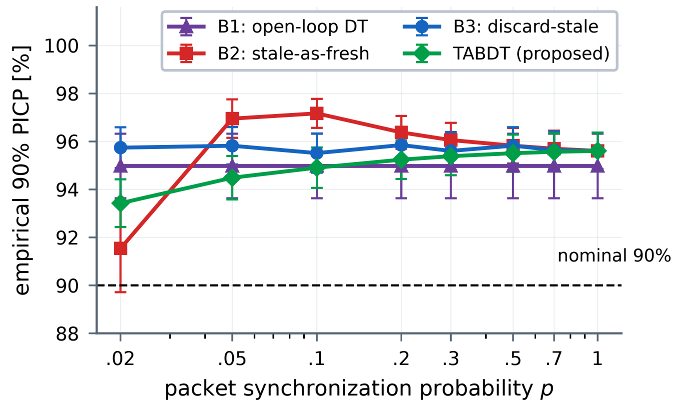

Fig. 4. Empirical coverage $\mathrm{PICP}_{0.9}$ on a log abscissa. Bars are defined in Section VI-B; the dashed line is the nominal 90% target. TABDT covers 93.4–95.6%.

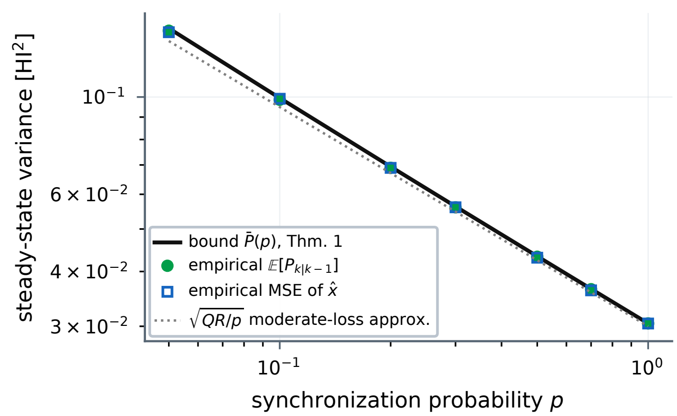

Fig. 5. Validation of Theorem 1. The closed-form bound $\bar{P}(p)$ exceeds the empirical prior variance and MSE at every evaluated p, with at most 1.3% slack over this range, and follows the $\sqrt{QR/p}$ moderate-loss approximation.

3) Prediction-interval coverage: Calibration is distinct from point accuracy [8]. Figure 4 shows conservative synthetic intervals: TABDT covers 93.4–95.6%, B3 95.5–95.9%, and B2 91.5% at $p = 0.02$ but 95.6–97.2% elsewhere. Thus the RMSE gain alone does not establish calibration.

4) Tightness of the analytical bound: Across 4 000 anchor paths in Fig. 5, $\bar{P}(p)/\mathbb{E}[P]$ ranges from 1.000 at $p = 1$ to 1.013 at $p = 0.05$. The $\sqrt{QR/p}$ curve has the moderate-loss variance slope $-1/2$; below $p \approx Q/R$, (21) instead gives the $p \to 0$ slope -1. Proposition 1 uses the closed-form bound, not this moderate-regime approximation.

5) Sensitivity and external age-compensation test: The pre-specified η/η^* sweep changes RMSE by at most 6.5%; at $p = 0.05$, ratios 1 and 100 give 54.7 and 51.2 h. This tests robustness, not empirical optimality of η^*.

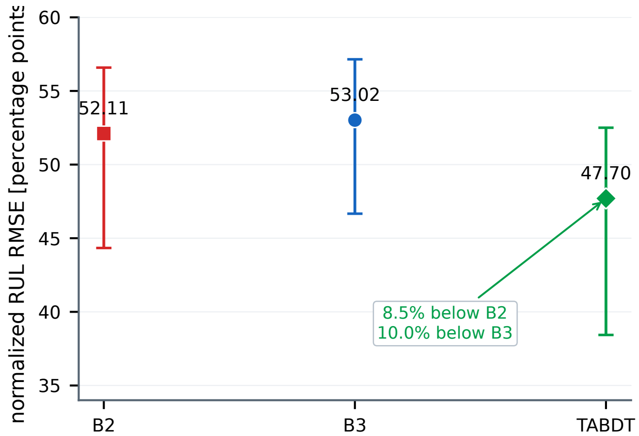

Fig. 6. XJTU-SY age-compensation test at $p = 0.02$. Bars are bearing-clustered 95% intervals; network repetitions are not independent specimens.

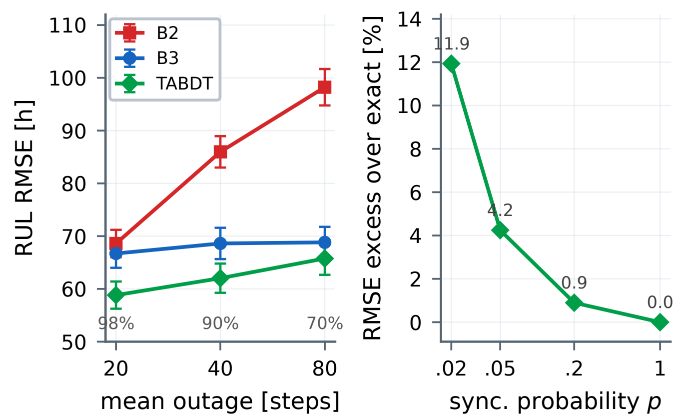

Fig. 7. Stress tests. Left: Markov-link RMSE at timely fraction 0.05 (descriptive bars as in Section VI-B; delivered fractions below). Right: TABDT excess over exact replay on a common 25-step grid.

A single-channel XJTU-SY test uses all 15 bearing lives under three operating conditions (9 216 one-minute records; 32 768 samples per record) [22]. Although two channels are recorded, the proxy uses only horizontal-vibration peak amplitude h: $\mathrm{clip}[(h/b-1)/9, 0, 1.2]$, which maps condition baseline b to zero and $10b$ to one. This is a normalization, not a failure label. Within each condition, leave-one-bearing-out calibration uses $m = 4$ lives and residual variance $s_w^2 + (1 + 1/m)s_b^2$, where s_w^2 is the mean within-bearing residual variance and s_b^2 is the variance of bearing-level residual means. The factor 1.25 is the predictive adjustment for a new bearing relative to the calibration mean, not a coverage-tuned value. Forty common packet realizations per bearing use $p = 0.02$ and a 30-record deadline. The primary aggregate includes all 15 lives, while the prespecified $\geq$ 100-record cohort of 13 lives is retained as a sensitivity analysis. Between packet arrivals, TABDT propagates the state using the calibrated drift; B2 and B3 hold their last accepted value.

As shown in Fig. 6, TABDT achieves a normalized RUL RMSE of 47.70 points, compared with 52.11 for stale-as-fresh fusion and 53.02 for discard-stale filtering. These values correspond to reductions of 8.5% and 10.0%, respectively. Bearing-clustered paired 95% bootstrap intervals for the absolute reductions are 3.59–6.72 and 4.12–9.56 points, both excluding zero. The 13-life sensitivity analysis yields 47.77, 52.16, and 53.01 points, with 76.7% coverage. Across all 15 lives, the nominal 90% interval covers only 76.9%; varying the between-bearing factor from 1 to 2 raises coverage from 72.4% to 84.5%. The point predictions support age compensation, but the interval results are insufficient to claim calibration for field use.

6) Stress tests beyond the model assumptions: Figure 7 reports two tests on the same 400-unit population. First, a two-state Markov link fixes the timely fraction at 0.05 while the mean outage length increases from 20 to 80 steps. TABDT retains the observed method ordering under these bursts, although the independent-delay certificate does not cover this case [20]; the result is therefore empirical. Second, retrospective reprocessing of every received measurement provides a reference posterior for assessing the forward approximation. TABDT's RMSE exceeds this reference by 11.9% at $p = 0.02$, 0.9% at $p = 0.2$, and 0% at $p = 1$. Exact reprocessing requires $O(tn^3 + Stn^2)$ work per evaluation at time t ($O(St)$ for fixed $n = 2$), whereas TABDT retains the online complexity reported above.

7) Network-aware maintenance decisions: Figure 8 translates the network-dependent certificate into the maintenance surrogate (23). Analytical curves are shown for three network conditions and compared with separate Monte Carlo policy simulations of 1 500 known-drift, single-sensor anchor paths per point. For the sampled points with $\xi \leq 9.0$, the two cost rates agree within 0.6%. The approximation deteriorates near D, as expected because the surrogate neglects diffusion overshoot: at $(p, \xi) = (0.2, 9.5)$, the simulated rate is 7.3% higher. These sampled points show that overshoot matters near D but do not establish a general safety-margin cutoff.

Within its stated conditions, the optimum moves earlier as the network deteriorates. Specifically, ξ^* is 9.42 at $p = 0.9$, 9.34 at $p = 0.5$, and 9.18 at $p = 0.2$. The corresponding safety margin widens from 0.58 to 0.82 HI, while the minimum surrogate cost increases from 2.135 to 2.197×10^{-3} per hour. Both sufficient conditions in Proposition 1 hold at these operating points; the smallest value of $(r^*)^2\xi^*$ is 88.5, compared with $D = 10$.

D. Threats to Validity and Limitations

Three limitations delimit the claims made here. First, Theorem 1 concerns a linear Gaussian, known-drift, single-sensor, fresh-only anchor with independent packet arrivals. The Markov-outage experiment examines correlated delivery only empirically [20]. Second, the main study is synthetic and uses fixed trust scores. The XJTU-SY experiment evaluates the packet-use rule on a single peak-amplitude proxy from 15 accelerated bearing lives; it is not a full validation of the degradation model. Third, TABDT's forward OOSM correction omits correlations retained by exact retrodiction [17], [18]. Its measured error relative to exact reprocessing is empirical, and the maintenance expression (23) remains a surrogate. The external interval coverage of 76.9% reinforces these limits. Nonlinear degradation, online trust estimation, correlated fading, implementation profiling, and plant trials remain necessary before deployment.

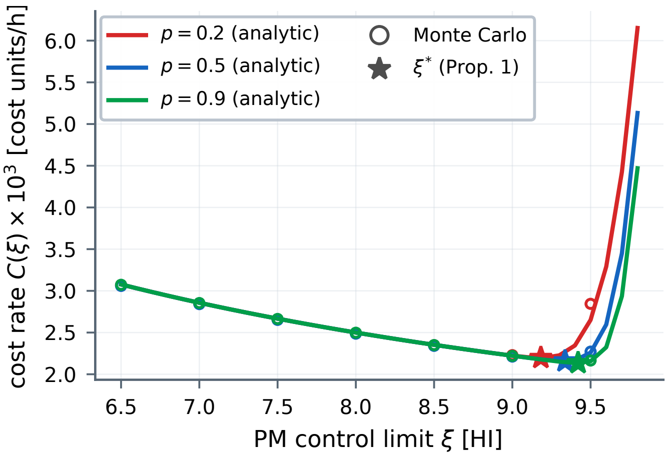

Fig. 8. Long-run cost rate against PM limit ξ: (23), 1 500-path anchor simulations, and ξ^*. Under Proposition 1's conditions, poorer synchronization moves ξ^* earlier.

VII. Conclusion

TABDT combines age compensation and sensor-specific precision within a Bayesian degradation twin, then uses a network-dependent variance certificate to inform the preventive-maintenance limit. In a 4 000-path anchor experiment, the bound is within 1.3% of empirical variance. In the separate 400-unit study, TABDT reduces RUL RMSE by as much as 27.9%, and the nominal intervals cover 93.4–95.6% of observations. Across 15 XJTU-SY bearing lives, normalized RMSE falls to 47.70 points from 52.11 and 53.02 for the two delayed-packet baselines. The corresponding coverage of 76.9% identifies interval calibration under measured degradation as the principal requirement before plant deployment.

References

[1] R. Liu, Q. Zhang, T. Han, B. Yang, W. Zhang, S. Yin, and D. Zhou, "Survey on foundation models for prognostics and health management in industrial cyber-physical systems," *IEEE Trans. Ind. Cyber-Phys. Syst.*, vol. 2, pp. 264–280, 2024, doi: 10.1109/TICPS.2024.3425326.

[2] J. Zhang, J. Tian, H. Luo, S. Wu, S. Yin, and O. Kaynak, "Prognostics for the sustainability of industrial cyber-physical systems: From an artificial intelligence perspective," *IEEE Trans. Ind. Cyber-Phys. Syst.*, vol. 2, pp. 495–507, 2024, doi: 10.1109/TICPS.2024.3433492.

[3] F. Tao, H. Zhang, A. Liu, and A. Y. C. Nee, "Digital twin in industry: State-of-the-art," *IEEE Trans. Ind. Informat.*, vol. 15, no. 4, pp. 2405–2415, Apr. 2019, doi: 10.1109/TII.2018.2873186.

[4] M. S. Azari, L. Ricci, S. Santini, and F. Flammini, "An intelligent diagnostic framework based on digital twins and partial transfer learning: Methodology and industrial application," *IEEE Trans. Ind. Cyber-Phys. Syst.*, vol. 3, pp. 1–13, 2025, doi: 10.1109/TICPS.2024.3490502.

[5] A. Kumar and N. Kumar, "State-space driven digital twin for condition monitoring and predictive health assessment in grid-integrated power converter system," *IEEE Trans. Ind. Cyber-Phys. Syst.*, vol. 3, pp. 464–471, 2025, doi: 10.1109/TICPS.2025.3586823.

[6] Y. Al-Karawi, H. S. Mogheer, K. K. Hasan, and A. H. Dawwd, "Digital twin and native-AI empowered 6G networks: Architectures, applications, evaluation, and open challenges," in *Proc. Int. Russian Smart Industry Conf. (SmartIndustryCon)*, 2026, pp. 13–20, doi: 10.1109/SmartIndustryCon68821.2026.11493034.

[7] E. Tabassi, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST AI 100-1, National Institute of Standards and Technology, Gaithersburg, MD, USA, 2023, doi: 10.6028/NIST.AI.100-1.

[8] Z.-Q. Ding, Q. Qin, Y.-F. Zhang, and Y.-H. Lin, "An uncertainty quantification and calibration framework for RUL prediction and accuracy improvement," *IEEE Trans. Instrum. Meas.*, vol. 73, Art. no. 2535613, pp. 1–13, 2024, doi: 10.1109/TIM.2024.3485392.

[9] B. Sinopoli, L. Schenato, M. Franceschetti, K. Poolla, M. I. Jordan, and S. S. Sastry, "Kalman filtering with intermittent observations," *IEEE Trans. Autom. Control*, vol. 49, no. 9, pp. 1453–1464, Sep. 2004, doi: 10.1109/TAC.2004.834121.

[10] R. D. Yates, Y. Sun, D. R. Brown, S. K. Kaul, E. Modiano, and S. Ulukus, "Age of information: An introduction and survey," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 5, pp. 1183–1210, May 2021, doi: 10.1109/JSAC.2021.3065072.

[11] N. Ahmad, M. Egan, J.-M. Gorce, J. S. Dibangoye, and F. Le Mouël, "Codesigned communication and data analytics for condition-based maintenance in smart buildings," *IEEE Internet Things J.*, vol. 10, no. 18, pp. 15847–15856, Sep. 2023, doi: 10.1109/JIOT.2023.3266029.

[12] T. Yang and C. Lv, "A secure sensor fusion framework for connected and automated vehicles under sensor attacks," *IEEE Internet Things J.*, vol. 9, no. 22, pp. 22357–22365, Nov. 2022, doi: 10.1109/JIOT.2021.3101502.

[13] K. Feng, Y. Xu, Y. Wang, S. Li, Q. Jiang, B. Sun, J. Zheng, and Q. Ni, "Digital twin enabled domain adversarial graph networks for bearing fault diagnosis," *IEEE Trans. Ind. Cyber-Phys. Syst.*, vol. 1, pp. 113–122, 2023, doi: 10.1109/TICPS.2023.3298879.

[14] X.-S. Si, W. Wang, C.-H. Hu, and D.-H. Zhou, "Estimating remaining useful life with three-source variability in degradation modeling," *IEEE Trans. Rel.*, vol. 63, no. 1, pp. 167–190, Mar. 2014, doi: 10.1109/TR.2014.2299151.

[15] Z. Huang, Z. Xu, W. Wang, and Y. Sun, "Remaining useful life prediction for a nonlinear heterogeneous Wiener process model with an adaptive drift," *IEEE Trans. Rel.*, vol. 64, no. 2, pp. 687–700, Jun. 2015, doi: 10.1109/TR.2015.2403433.

[16] Q. Zhai and Z.-S. Ye, "RUL prediction of deteriorating products using an adaptive Wiener process model," *IEEE Trans. Ind. Informat.*, vol. 13, no. 6, pp. 2911–2921, Dec. 2017, doi: 10.1109/TII.2017.2684821.

[17] Y. Bar-Shalom, "Update with out-of-sequence measurements in tracking: Exact solution," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 38, no. 3, pp. 769–777, Jul. 2002, doi: 10.1109/TAES.2002.1039398.

[18] Y. Bar-Shalom, H. Chen, and M. Mallick, "One-step solution for the multistep out-of-sequence-measurement problem in tracking," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 40, no. 1, pp. 27–37, Jan. 2004, doi: 10.1109/TAES.2004.1292140.

[19] M. Greiff and K. Berntorp, "Asynchronous variational-Bayes Kalman filtering," in *Proc. 63rd IEEE Conf. Decision Control (CDC)*, Milan, Italy, 2024, pp. 1987–1992, doi: 10.1109/CDC56724.2024.10886035.

[20] D. E. Quevedo, A. Ahlén, and K. H. Johansson, "State estimation over sensor networks with correlated wireless fading channels," *IEEE Trans. Autom. Control*, vol. 58, no. 3, pp. 581–593, Mar. 2013, doi: 10.1109/TAC.2012.2212515.

[21] L. Yang, Y. Chen, X. Ma, Q. Qiu, and R. Peng, "A prognosis-centered intelligent maintenance optimization framework under uncertain failure threshold," *IEEE Trans. Rel.*, vol. 73, no. 1, pp. 115–130, Mar. 2024, doi: 10.1109/TR.2023.3273082.

[22] B. Wang, Y. Lei, N. Li, and N. Li, "A hybrid prognostics approach for estimating remaining useful life of rolling element bearings," *IEEE Trans. Rel.*, vol. 69, no. 1, pp. 401–412, Mar. 2020, doi: 10.1109/TR.2018.2882682.